

CrimeInsight: Crime Data Pipeline and Business Intelligence System

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PROJECT OVERVIEW

CrimeInsight is a data engineering and business intelligence system that demonstrates the complete **Data Engineering Lifecycle** using crime data. The system collects crime records from multiple sources such as CSV files, open datasets, and simulated streaming sources.

The project uses **batch ingestion** to collect and store raw crime data in a raw data layer. The data is then validated, cleaned, deduplicated, normalized, and transformed into structured datasets. Data modeling, partitioning, schema consistency, data quality, scalability, and fault tolerance are considered during the pipeline.

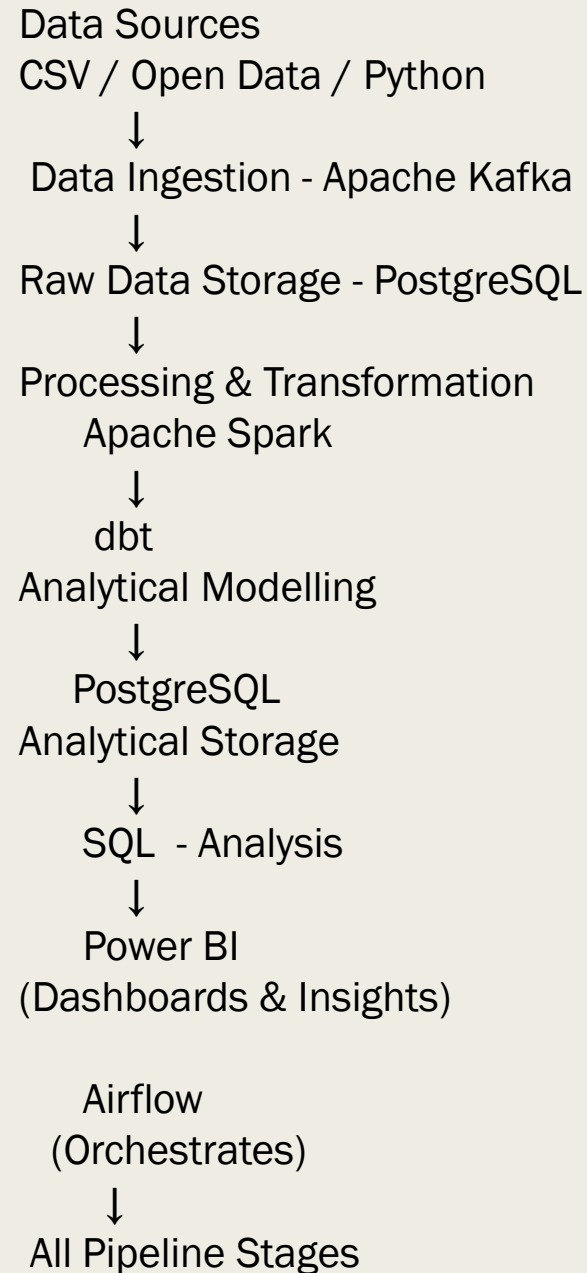
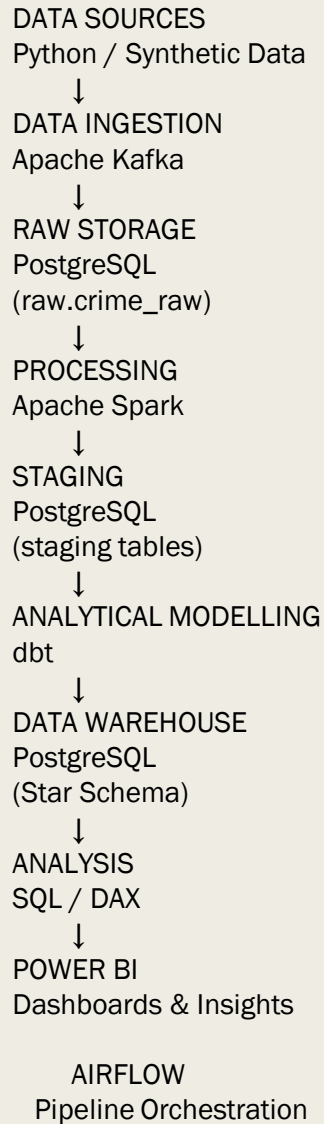
The processed data is stored in an **analytical storage layer** and used for business intelligence analysis. The system identifies crime patterns based on **crime frequency, location, time period, crime category, and trends**.

Dashboards and reports present these insights in an easy-to-understand format.

METHODOLOGY

- **Data Collection** – Collect crime records from CSV files, open datasets, and synthetically generated sources representing different crime attributes.
- **Data Ingestion** – Ingest historical crime records through batch processing and continuously generated records through simulated streaming.
- **Data Cleaning** – Identify and handle missing values, duplicate records, invalid entries, and inconsistent crime fields.
- **Data Transformation** – Standardize formats, normalize relevant fields, filter invalid records, and prepare consistent datasets for analysis.
- **Data Modeling** – Use dbt to create structured analytical models from the transformed crime data.
- **Data Storage** – Maintain processed and analytical datasets in a structured storage layer for efficient access and analysis.
- **Data Analysis** – Analyse crime records based on crime type, location, and time to determine frequencies, trends, distributions, and patterns.
- **Visualization** – Present analytical results through interactive dashboards, highlighting crime trends, frequently occurring categories, and geographical distributions.
- **Insight Generation** – Interpret the dashboard results to identify crime trends, hotspots, frequently occurring categories, and time-based patterns.
- **Workflow Automation** – Use Apache Airflow to schedule, coordinate, and monitor the different stages of the data pipeline.

SYSTEM DESIGN



ARCHITECTURE



CrimeInsight - Data Engineering & BI Architecture

USE OF MODERN TOOLS AND TECHNOLOGIES

Tool / Technology	Purpose
Python	Generate and prepare realistic crime data
CSV / Open Datasets	Provide crime data from different sources
Apache Kafka	Real-time crime data ingestion and streaming
PostgreSQL	Store raw and processed crime data
Apache Spark	Data cleaning, validation, transformation, and processing
dbt	Data modeling and analytical transformations
SQL	Query and analyze crime data
Apache Airflow	Schedule and orchestrate pipeline workflows
Power BI	Create interactive dashboards and visualizations
Docker	Containerize project components and maintain a consistent environment

IMPLEMENTATION

- Developed Python scripts to generate synthetic crime records and stream them through the **crime-events Kafka topic** using a producer-consumer workflow.
- Implemented PostgreSQL schemas for storing incoming records in **raw.crime_raw** and prepared the **staging.crime_clean** table for processed data. The raw layer has been verified with **275+ records**.
- Developed **PySpark scripts** for cleaning, validating, deduplicating, enriching, and transforming crime data before analytical processing.
- Implemented **dbt models** including stg_crime, fact_crime, dim_date, dim_location, and dim_crime_type, along with data-quality tests.
- Developed an **Apache Airflow DAG** to coordinate data generation, Kafka ingestion, Spark processing, and dbt execution on a scheduled workflow.
- Configured **Docker Compose** to run PostgreSQL, Kafka, Zookeeper, and Airflow together in a consistent environment.
- The data-engineering components are coded, while final Spark execution, dbt execution, and Power BI dashboard creation/connection remain to be completed.

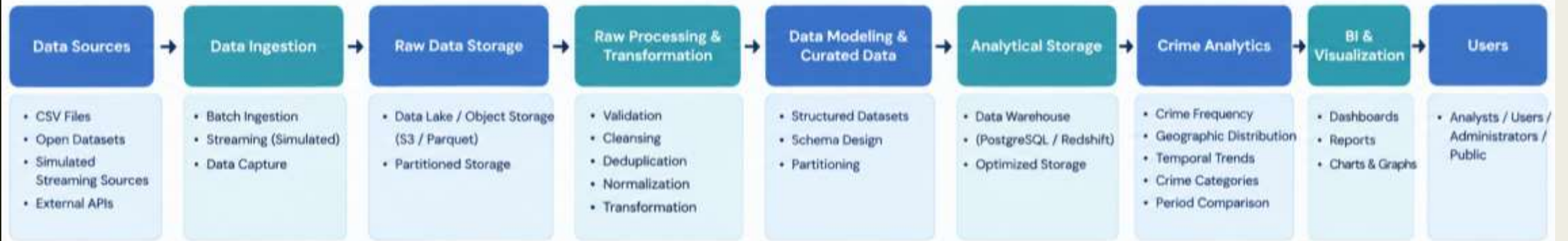
Implementation Flow

Python / CSV / Open Data → Kafka → PostgreSQL → Spark → dbt → Power BI

CODING QUALITY

- **Modular Code:** Pipeline is divided into separate components for ingestion, cleaning, transformation, and analysis.
- **Readable & Maintainable:** Clear naming conventions and well-structured Python and Spark code.
- **Data Validation:** Checks for missing, duplicate, invalid, and inconsistent records.
- **Error Handling:** Handles invalid data and pipeline failures without affecting the complete workflow.
- **Reusability:** Functions and processing modules can be reused for different crime datasets.
- **Scalability:** Spark-based processing supports large volumes of crime data.
- **Automation:** Airflow reduces manual execution and manages pipeline dependencies.
- **Consistency:** Docker provides a consistent environment for development and execution.
- **Testing & Reliability:** Data quality checks ensure that processed datasets are accurate and analysis-ready.

PROJECT WORKFLOW



Key Data Engineering Considerations

» DAG crimeinsight_pipeline

△ Details **■ Graph** ▢ Gantt <> Code

Layout:

Left -> Right ▾



FUNCTIONALITY

- **Crime Data Generation:** Generates realistic crime records using Python.
- **Real-Time Data Ingestion:** Continuously receives crime events using Kafka.
- **Data Quality Management:** Detects and handles missing, duplicate, and inconsistent records.
- **Data Processing:** Cleans, filters, transforms, and aggregates crime data using Spark.
- **Data Modeling:** Converts processed data into structured analytical datasets using dbt.
- **Data Storage:** Maintains raw and processed crime data in PostgreSQL.
- **Automated Pipeline:** Airflow schedules and manages the complete data workflow.
- **Crime Analysis:** Identifies patterns based on crime type, location, and time.
- **Interactive Visualization:** Power BI provides dashboards, charts, and reports.
- **Insight Generation:** Helps identify crime trends, hotspots, and frequently occurring crime categories.

THANK YOU